FORMATIVE ASSESSMENT-03

MODEL QUESTION PAPER

CLASS: 10	MARKS: 20	MATHEMATICS

I. Choose the correct answer for given below ---------- 2x1=2

1. The degree of the polynomial p (x) = $3x^2+2x-4$ is

A) 2 B) 3 C) 4 D) 1

2. The value of $\sin 30^0+\cos 60^0$ is…….

A) $\frac{1}{2}$ B) $\frac{3}{2}$ C) $\frac{1}{4}$ D)1

II. Answer the following questions ---------- 2x1=2

3. Write the discriminant of $ax^2+bx+c=0$.

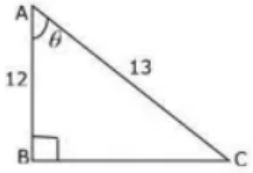
4. Find the sum of the zeroes of the polynomial $P(x) = 2x^2-5x+6$.

III. Answer the following questions ---------- 3x2=6

5. Find the quotient and the remainder when p(x) $=x^2+3x-18$ is divided by g (x) =x-6.

6. Find the roots of the quadratic equation $2x^2 + 4x - 5 = 0$ using the 'quadratic formula'.

7. In the figure given below find the value of $\sin\theta$ and $\text{cosec}\,\theta$.



IV. Answer the following questions---------- 2x3=6

8. A two digit number is such that the product of the digits is 14. When 45 is added to the number then digits are interchange their places. Find the number.

OR

An express train makes a run of 240 Km at a certain speed. Another train whose speed is 12 Km/h less takes an hour longer to cover the same distance. Find the speed of express train in Km/h.

9. Prove that $\frac{sin\theta}{1-cot\theta}+\frac{cos\theta}{1-tan\theta}$ = sinϴ+cosϴ.

V. Answer the following questions ---------- 1x4=4

10. The angle of elevation of the top of an unfinished vertical building on a ground, at a point which is 100m from the base of the building is 45^0. How much height the building must be raised so that its angle of elevation from the same point be 60^0?.

OR

Froom a point on the ground the angle of elevation of the top and bottom of a transmission tower fixed at the top of a 20m high building are 600 and 450 respectively. Find the height of the transmission tower.

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